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P2 — Forces

OCR GCSE Physics A (J249) — Gateway Science · Paper 2

P2.1 Scalars and Vectors

A **scalar** has size (magnitude) only. A **vector** has both magnitude and direction. Arrows on diagrams represent vectors — length shows size, direction of the arrow shows direction of the quantity.

Scalar quantities	Vector quantities
Distance, speed, mass, time, energy, temperature	Displacement, velocity, acceleration, force, momentum, weight

Notice the pairs: **distance / displacement** and **speed / velocity**. Walking 5 m north then 5 m south gives a distance of 10 m, but the displacement (straight-line from start to finish) is 0 m. The direction matters.

P2.2 Contact and Non-contact Forces

A **force** is a push or a pull on an object, measured in **newtons (N)**. Forces are vectors. They can be **contact** (objects must touch) or **non-contact** (forces act at a distance through fields).

Contact forces	Non-contact forces
Friction, air resistance, drag, tension, normal contact force, upthrust, applied/push force	Gravitational, electrostatic, magnetic

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Weight vs Mass

Mass is the amount of matter in an object — measured in kg; unchanged anywhere in the universe.
Weight is the *force* of gravity on an object — measured in N; changes depending on gravitational field strength.

$$\mathbf{W = m \times g} \quad (\text{N}) = (\text{kg}) \times (\text{N/kg})$$

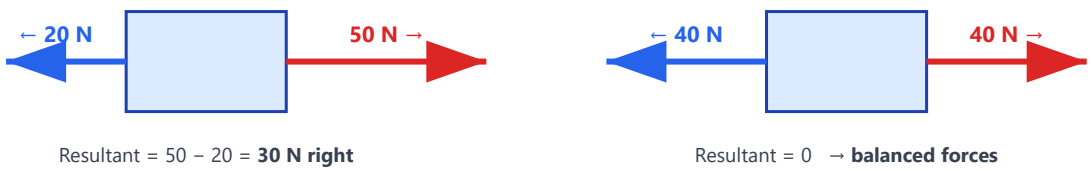
On Earth $g \approx 9.8$ N/kg (often rounded to 10). On the Moon $g \approx 1.6$ N/kg.

Worked Example: An astronaut has mass 80 kg. Their weight on Earth is 80 × 9.8 = **784 N**. Their weight on the Moon is 80 × 1.6 = **128 N**. Mass is unchanged at 80 kg everywhere.

Resultant Force

When several forces act on an object, the **resultant force** is the single force that would have the same effect as all of them combined. If forces are along a straight line, just add or subtract them; forces in the same direction add, opposing forces subtract.

Adding forces along a line



Balanced forces: object is either stationary or moving at constant velocity.

Perpendicular Forces — Scale Diagrams

When two forces act at right angles to each other, draw a scale diagram (tip-to-tail) and use a ruler + protractor to measure the resultant. You can also use Pythagoras: the magnitude of the resultant of two perpendicular forces of size a and b is $\sqrt{(a^2 + b^2)}$.

P2.3 Work Done

Work done is the energy transferred when a force moves an object. 1 joule = 1 newton × 1 metre. So doing 1 J of work is the same as transferring 1 J of energy.

$$\mathbf{W = F \times d} \quad (\mathrm{J}) = (\mathrm{N}) \times (\mathrm{m})$$

Worked Example: A horse pulls a cart with a force of 400 N for 25 m. Work done = 400 × 25 = **10 000 J**. This energy is transferred mostly to kinetic energy of the cart, plus some to heat via friction.

Work done against **friction** heats up the surfaces in contact. That's why rubbing your hands warms them and why car brakes get hot during braking.

P2.4 Motion — Describing How Things Move

Speed, Velocity, Acceleration

$$\textbf{speed = distance ÷ time} \quad (\mathrm{m/s}) = (\mathrm{m}) / (\mathrm{s})$$

$$\textbf{acceleration = change in velocity ÷ time} \quad a = (v - u) / t \quad (\mathrm{m/s^2})$$

Typical speeds (m/s)	Value
Walking	1.5
Running	3
Cycling	6
Car (urban)	13 (≈ 30 mph)

Train	55
Sound in air	330
Wind (breeze → storm)	5 → 30

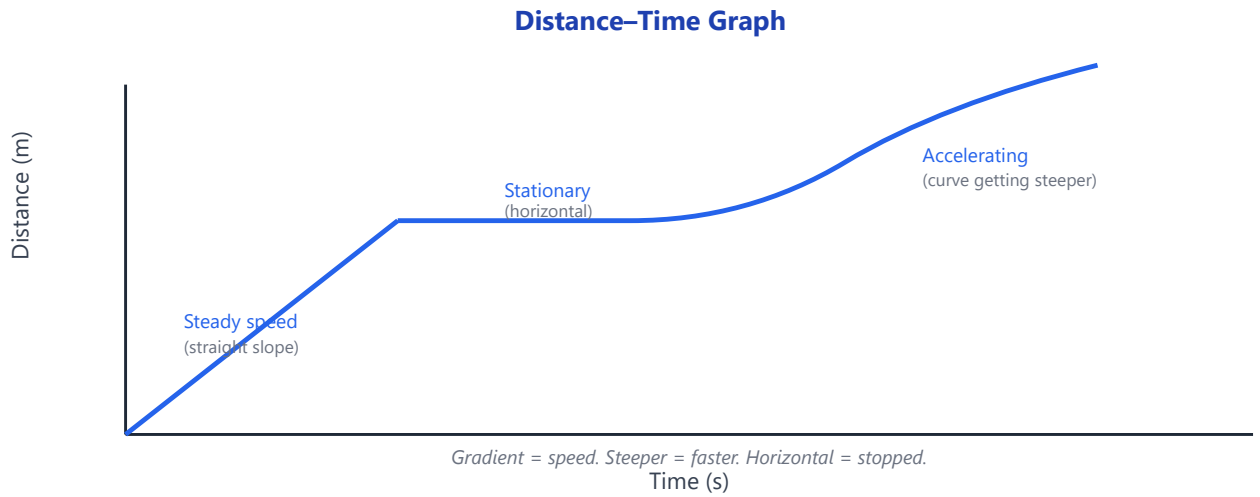
Uniform Acceleration Equation

$$v^2 = u^2 + 2 \times a \times s$$

v = final velocity (m/s), u = initial velocity (m/s), a = acceleration (m/s²), s = distance (m)

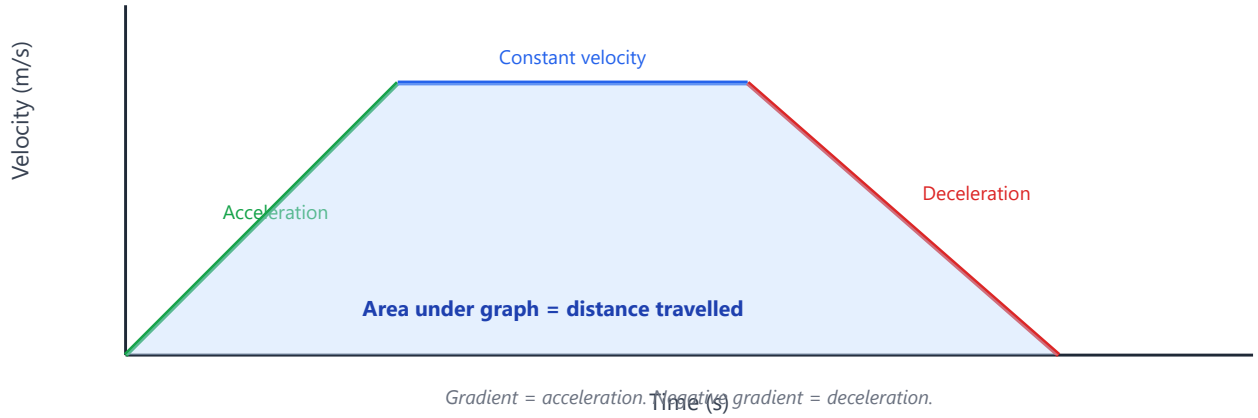
Worked Example: A car accelerates from rest at 2 m/s² over a distance of 50 m. Final velocity? $v^2 = u^2 + 2as = 0 + 2 \times 2 \times 50 = 200$. So $v = \sqrt{200}$ = **14.1 m/s**.

Distance-Time Graphs



Velocity-Time Graphs

Velocity–Time Graph



Gradient = acceleration. Negative gradient = deceleration.

Exam Tip: Do not confuse distance-time and velocity-time graphs.
• Distance-time: gradient = speed, horizontal line = stationary.
• Velocity-time: gradient = acceleration, horizontal line = constant velocity, area under graph = distance.

P2.5 Newton's Laws of Motion

Newton's First Law — Inertia

> An object continues at rest or at constant velocity unless acted on by a resultant force. Another way: if forces are balanced, velocity doesn't change.

This is why seatbelts matter: during a sudden stop, your body continues moving forward until an external force (the belt or the windshield) stops it. The tendency to keep doing what you were doing is called **inertia**.

Newton's Second Law

$$\mathbf{F = m \times a} \quad (\mathrm{N}) = (\mathrm{kg}) \times (\mathrm{m/s^2})$$

A resultant force on a mass causes an acceleration in the same direction.

Worked Example: A car of mass 1200 kg accelerates at 3 m/s^2. Resultant force needed? F = ma = 1200 × 3 = **3600 N**.
If the same 3600 N acted on a lighter car of mass 800 kg, acceleration would be 3600/800 = **4.5 m/s^2**. Lighter mass → bigger acceleration for the same force.

Inertial mass is defined by F = ma: it is the property that resists a change in motion. A heavier mass needs a bigger force to produce the same acceleration.

Newton's Third Law

When two objects interact, the forces they exert on each other are **equal in size and opposite in direction**. Every action has an equal and opposite reaction.

Examples:

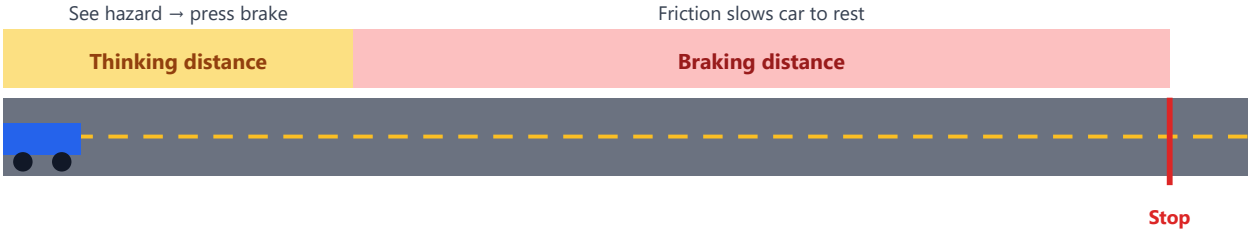
- You push down on a chair; the chair pushes up on you with the same force (normal contact force).
- A rocket pushes exhaust gas downwards; the gas pushes the rocket upwards with equal force.
- Earth pulls you down with your weight; you pull Earth up with the same force (Earth is far too massive to notice).

Exam Tip: The action and reaction act on *different* objects. This is easy to confuse with balanced forces (which act on the *same* object). A ball on a table: the weight (Earth pulls ball) and normal contact force (table pushes ball) are balanced forces on the ball — NOT a Newton's-third-law pair.

P2.6 Stopping Distances

Stopping distance = **thinking distance** + **braking distance**.
Thinking distance = distance travelled while the driver reacts (during reaction time).
Braking distance = distance travelled while the brakes slow the car to rest.

Stopping Distance = Thinking + Braking



Factors Affecting Thinking Distance

Thinking distance depends on the driver's **reaction time** (≈ 0.2 – 0.9 s) and the vehicle's **speed**:

- Alcohol, drugs, tiredness — slow reactions, longer thinking distance.
- Using a phone / distractions — same effect.
- Higher speed — more distance covered in the same reaction time.

Factors Affecting Braking Distance

- **Speed** — braking distance is proportional to $\mathbf{v^2}$. Doubling speed quadruples braking distance.
- **Road surface** — wet, icy, oily or gravelled roads reduce friction, increase braking distance.
- **Tyre / brake condition** — worn tyres or worn brake pads are less effective.
- **Mass of vehicle** — heavier vehicles (loaded lorry) need more force to decelerate, so longer braking distance.

Speed (mph)	Thinking (m)	Braking (m)	Total stopping (m)
20	6	6	12
30	9	14	23
40	12	24	36
50	15	38	53
60	18	55	73
70	21	75	96

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Hard Braking — Dangers

Bringing a fast-moving car to rest requires a **large deceleration**, which needs a **large force** (F = ma).

- The brakes do a huge amount of work in a short distance — they heat up dramatically.
- If brakes overheat, the friction pads lose grip (**brake fade**).
- Large deceleration can cause loss of control, especially if wheels skid on a low-friction surface.

P2.7 Momentum

Momentum (p) is a measure of how much motion an object has. It is mass × velocity. Momentum is a **vector**. Units: kg m/s.

$$\mathbf{p} = \mathbf{m} \times \mathbf{v} \quad (\text{kg m/s}) = (\text{kg}) \times (\text{m/s})$$

Conservation of Momentum

In a closed system (no external forces) the total momentum before an event equals the total momentum after. This is called **conservation of momentum**.

Force and Momentum Change

A force changes momentum. The **larger the force** or **longer the time** the force acts, the greater the change in momentum.

$$F = (mv - mu) / t \quad \text{(alternative form of } F = ma\text{)}$$

where (mv − mu) is the change in momentum.

Why Safety Features Work

Crumple zones, airbags, seatbelts and helmets all work on the same principle: they **increase the time** over which a person's momentum is reduced to zero, which **reduces the force** on them.

Safety feature	How it works
Seatbelt	Stretches slightly — increases stopping time, reduces force on chest.
Airbag	Inflates between driver and dashboard — longer deceleration, lower peak force.
Crumple zone	Front/rear of car crushes — absorbs energy, extends collision time.
Cycle helmet	Foam liner crushes on impact — extends time, spreads force over head.
Crash mats	Foam compresses — increases landing time, reduces impact force.

Worked Example: A 70 kg driver in a crash comes to rest from 15 m/s in 0.1 s (no seatbelt) or 0.5 s (with seatbelt + airbag).
Change in momentum = 70 × 15 = 1050 kg m/s.
• No restraint: F = 1050 / 0.1 = **10 500 N**.
• With restraint: F = 1050 / 0.5 = **2 100 N**.
Five times longer stopping time → five times less force. That can be the difference between life and death.

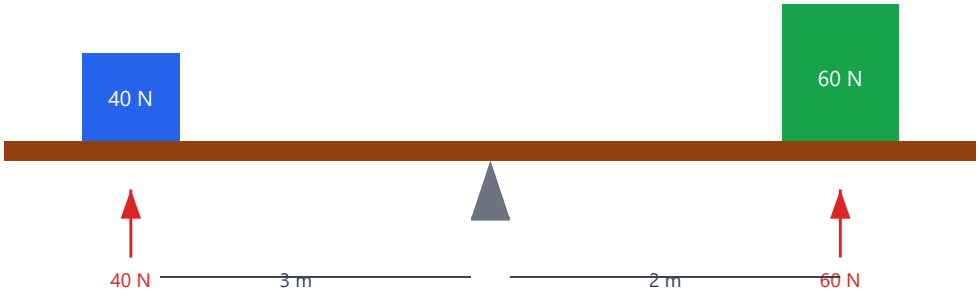
P2.8 Moments — Turning Effects

The **moment** of a force is its turning effect about a pivot. A longer lever gives a bigger turning effect for the same force. Unit: N m.

$$\textbf{Moment} = \mathbf{F} \times \mathbf{d} \quad (\text{N m}) = (\text{N}) \times (\text{m})$$

d is the *perpendicular* distance from pivot to line of action.

Principle of Moments — Balanced Seesaw



40 × 3 = 120 N m = 60 × 2 = 120 N m → balanced

Principle of Moments

For a balanced object: **total clockwise moment = total anticlockwise moment**.

> **Worked Example:** A child weighing 200 N sits 2.0 m from a pivot. How far from the pivot must a 160 N child sit on the other side for balance?
> Clockwise = anticlockwise → 200 × 2.0 = 160 × d → d = 400 / 160 = **2.5 m**.

Levers and Gears

Both are **force multipliers**: they let a small input force produce a large output force, at the cost of having to move a larger distance.

- **Levers** (spanners, crowbars, wheelbarrows): a long handle with a short load distance multiplies force.
- **Gears** interlocking wheels: a small driver gear turning a large driven gear transmits a larger turning force (but at slower speed).

Quick Summary — Key Equations for P2

Quantity	Equation	Units

Weight	$W = m\,g$	N
Work done	$W = F\,d$	J
Speed	$v = d / t$	m/s
Acceleration	$a = (v - u) / t$	m/s^2
Newton's 2nd law	$F = m\,a$	N
Uniform accel	$v^2 = u^2 + 2\,a\,s$	—
Momentum	$p = m\,v$	kg m/s
Force and momentum	$F = (mv - mu)/t$	N
Moment	$M = F\,d$	N m